
ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology

Synopsis

on

AI Powered Coding Practice Platform (Mini-Leetcode)

1. INTRODUCTION

1.1. Overview

In the modern technology-driven world, programming skills have become essential for students and professionals who aspire to work in the software industry. Many companies evaluate candidates through coding interviews and competitive programming assessments. However, students often struggle to develop strong problem-solving abilities due to the lack of structured practice platforms and personalized feedback. Traditional learning methods focus mainly on theoretical knowledge, which is often insufficient for mastering programming skills. Students need practical environments where they can solve coding problems, analyze their mistakes, and continuously improve their logical thinking abilities. Online coding platforms such as LeetCode and HackerRank have demonstrated the effectiveness of interactive problem-solving environments for learning programming. These platforms provide coding challenges that help users practice algorithms, data structures, and logical problem solving. The proposed project aims to develop an AI Powered Coding Practice Platform (Mini-LeetCode) that allows users to practice programming problems in an interactive environment. The system will provide coding challenges, evaluate user solutions automatically, and track user performance. The platform will also integrate intelligent analysis to monitor user progress and recommend appropriate problems based on the user's skill level. This will create a personalized learning environment that helps users improve their programming abilities effectively.

1.2. Purpose

The primary purpose of this project is to design and implement an intelligent coding practice platform that enables students and programmers to improve their programming skills through structured problem-solving exercises.

The system aims to achieve the following objectives:

Automated Coding Practice Environment: To provide an interactive platform where users can practice programming problems and submit solutions.

Automated Code Evaluation: To evaluate user-submitted code automatically using predefined test cases and provide instant feedback.

Performance Analysis: To track user performance and analyze coding results to identify strengths and weaknesses.

Personalized Learning: To recommend coding problems according to the user's skill level and previous performance.

Skill Development: To help students improve logical thinking, algorithmic skills, and programming proficiency.

2.LITERATURE SURVEY

2.1. Existing Problem

In Many students face difficulties in developing strong programming skills due to limited access to structured coding practice environments. Traditional classroom teaching often focuses on theoretical concepts rather than practical problem solving. Existing coding practice platforms provide large collections of problems but may lack personalized feedback or adaptive learning mechanisms that guide beginners effectively. Additionally, many students struggle to identify the most appropriate problems to practice based on their skill level.

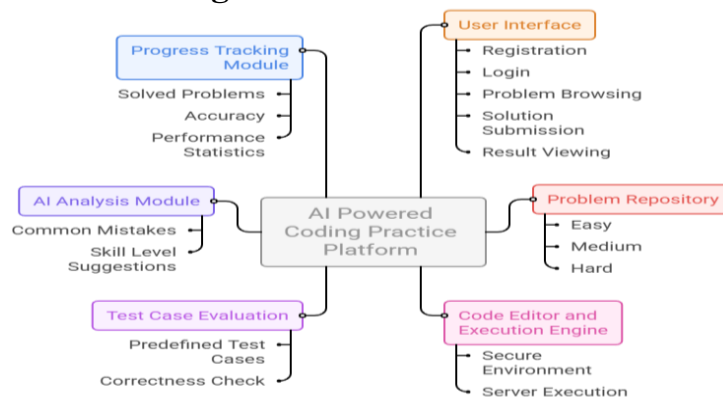
Manual evaluation of programming assignments can also be time-consuming for instructors, making it difficult to provide quick feedback to students. These limitations highlight the need for an intelligent system that can automatically evaluate coding solutions, track user progress, and provide personalized learning recommendations.

2.2. Proposed Solution

To The proposed system is an AI Powered Coding Practice Platform that provides an interactive environment for practicing programming problems. Users will be able to browse coding challenges categorized by difficulty level and topic. The platform will include a built-in code editor that allows users to write and submit their solutions directly through the interface. The submitted code will be executed in a secure environment and evaluated against predefined test cases. The system will determine whether the solution is correct and provide feedback to the user. The platform will also analyze user performance to track solved problems, accuracy, and progress over time. Based on this analysis, the system will recommend new problems that match the user's skill level. This intelligent learning environment will help users gradually improve their coding skills and prepare for technical interviews and competitive programming challenges.

3. THEORETICALANALYSIS

3.1. Block Diagram



The block diagram of the AI Powered Coding Practice Platform represents the flow of information from user interaction to performance analysis.

The system consists of the following components:

User Interface: This module allows users to interact with the platform. Users can register, log in, browse coding problems, submit solutions, and view their results.

Problem Repository: This module stores a large collection of coding problems categorized by difficulty levels such as easy, medium, and hard.

Code Editor and Execution Engine: This module allows users to write and run code directly in the browser. The submitted code is executed on the server using a secure code execution environment.

Test Case Evaluation: The system automatically checks user solutions against predefined test cases to determine whether the solution is correct.

AI Analysis Module: This module analyzes user performance, identifies common mistakes, and suggests appropriate practice problems based on the user's skill level.

Progress Tracking Module: The platform keeps track of solved problems, accuracy, and performance statistics to help users monitor their learning progress.

This architecture ensures efficient management of coding problems and provides a structured environment for learning programming skills.

3.2. Hardware/Software Designing

Hardware Requirements: The platform requires a server capable of handling user requests and executing code securely.

- The hardware requirements include:
- A computer server with a modern multi-core processor
- Minimum 8 GB RAM for handling code execution tasks
- Adequate storage space for storing problem datasets and user information
- Standard input/output devices for user interaction

Since the platform is web-based, users can access it through laptops, desktops, or mobile devices using a web browser.

Software Requirements:

The backend system will manage problem storage, code execution, user authentication, and performance analysis.

Frontend Development: The frontend interface will be developed using web technologies such as HTML, CSS, and JavaScript to provide an interactive coding environment.

Database Management: A database system will store user accounts, coding problems, submissions, and performance statistics.

Code Execution Environment: A secure code execution environment will run submitted programs and evaluate them against test cases.

Artificial Intelligence Integration: Artificial intelligence techniques will analyze user performance and recommend suitable coding problems to improve learning outcomes.

Deployment Platform: The system can be deployed on a cloud server to ensure scalability and availability for multiple users.

4. APPLICATIONS

The AI Powered Coding Practice Platform can be used in various educational and professional environments.

Programming Education: Students learning programming can use the platform to practice coding problems and improve their problem-solving skills.

Technical Interview Preparation: The platform can help job seekers prepare for coding interviews conducted by software companies.

Competitive Programming Training: The system can be used by competitive programmers to practice algorithmic problems and improve their coding speed.

University Programming Courses: Educational institutions can integrate the platform into programming courses to provide practical coding exercises.

Skill Development Programs: Coding bootcamps and training institutes can use the platform to evaluate and improve students' programming abilities.

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